Keras Feature Extraction

```
import tensorflow as tf
from tensorflow.keras.datasets import imdb
from tensorflow.keras.preprocessing.sequence import
                                    pad_sequences
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Embedding, SimpleRNN
import matplotlib.pyplot as plt
vocab_size = 10000
(X_train, y_train), (X_test, y_test) = imdb.load_data(num_words
                                    =vocab_size)
max_length = 200
X_train = pad_sequences(X_train, maxlen = max_length)
X_test = pad_sequences(X_test, maxlen = max_length)
model = Sequential()
model.add(Embedding(input_dim=vocab_size, output_dim
model.add(SimpleRNN(32))
model.add(Dense(1, activation='sigmoid'))
model.compile(optimizer='adam', loss='binary_cross_
            entropy', metrics=['accuracy'])
history = model.fit(X_train, y_train, epochs=5, batch_size=
            validation_split=0.2)
test_loss, test_acc = model.evaluate(X_test, y_test)
print("Test Accuracy:", test_acc)
plt.plot(history.history['accuracy'], label='Training Accura
plt.plot(history.history['val_accuracy'], label='Validation
                                    Accuracy')
plt.legend()
plt.title("Model Accuracy")
plt.xlabel("Epoch")
plt.ylabel("Accuracy")
plt.show()
```

OUTPUT

Epoch 1/5

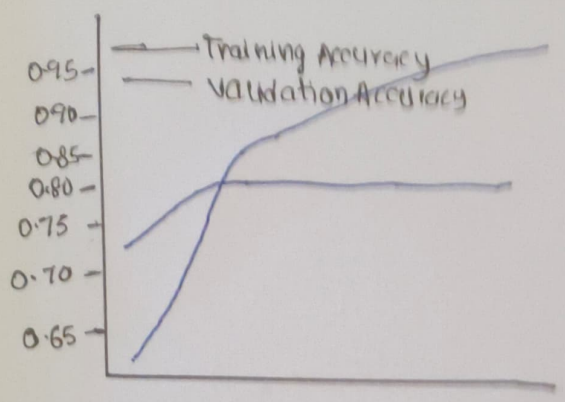
313/313 accuracy: 0.6708 loss: 0.5987 val_accuracy: 0.8888 val_loss: 0.4974

Epoch 2/5

313/313 accuracy: 0.8815 loss: 0.4075 val_accuracy: 0.8814 val_loss: 0.3986

Epoch 3/5

313/313 accuracy: 0.9029 loss: 0.2521 val_accuracy: 0.8854 val_loss: 0.3[illegible]

Epoch 4/5

313/313 accuracy: 0.9393 loss: 0.1703 val_accuracy: 0.8816 val_loss: 0.4[illegible]